भारतीय सांकेतिक विज्ञान संस्थ हैदराबाद
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Module 7: *MOSFET*

Long channel model and short channel effects

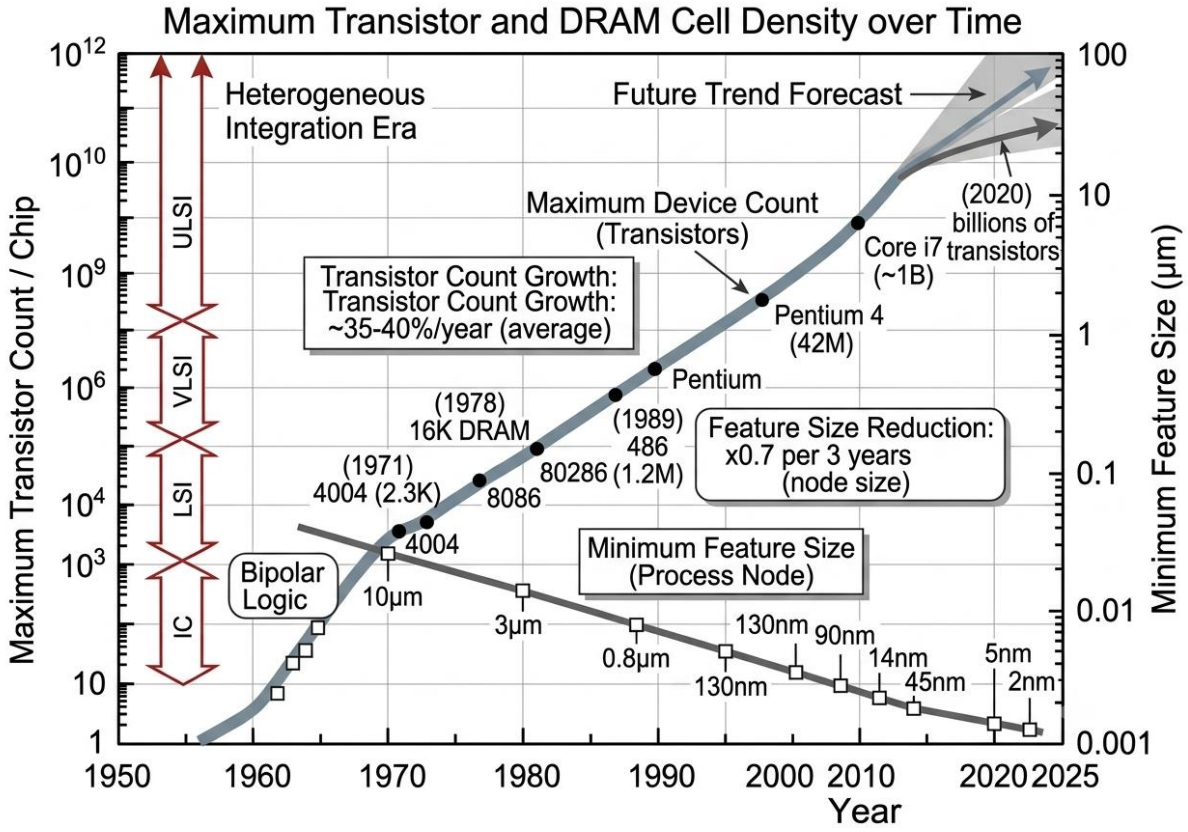
EE2104: Semiconductor Device Fundamentals

EE, IIT Hyderabad

Shubhadeep B (shubhadeep@ee at iith)

CMOS Scaling – From long to short channel

MOSFETs key invention 21st century: Computing revolution continues (AI era)

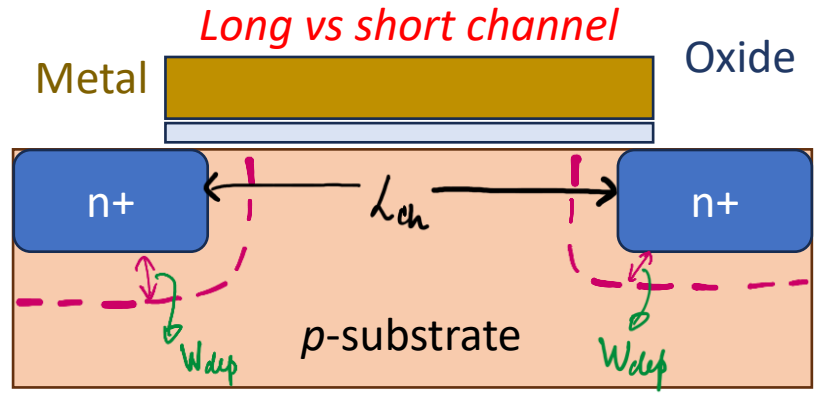


CMOS Scaling: Making transistors smaller cheaper and better performance:
Packing more punch in the same area

But has(is) not easy!

Learning Objectives:

- Long vs. short-channel MOSFET
- Operation of long-channel MOSFET *(band diagrams -> charge transport)*
- MOSFET Current in ON and OFF states
- Short-channel effects (DIBL, V_t roll-off)
- How do we continue to scale? <if time allows>



Simple definition:

Long channel: $L_{ch} >> W_{dep}$

Short channel: $L_{ch} \sim W_{dep}$

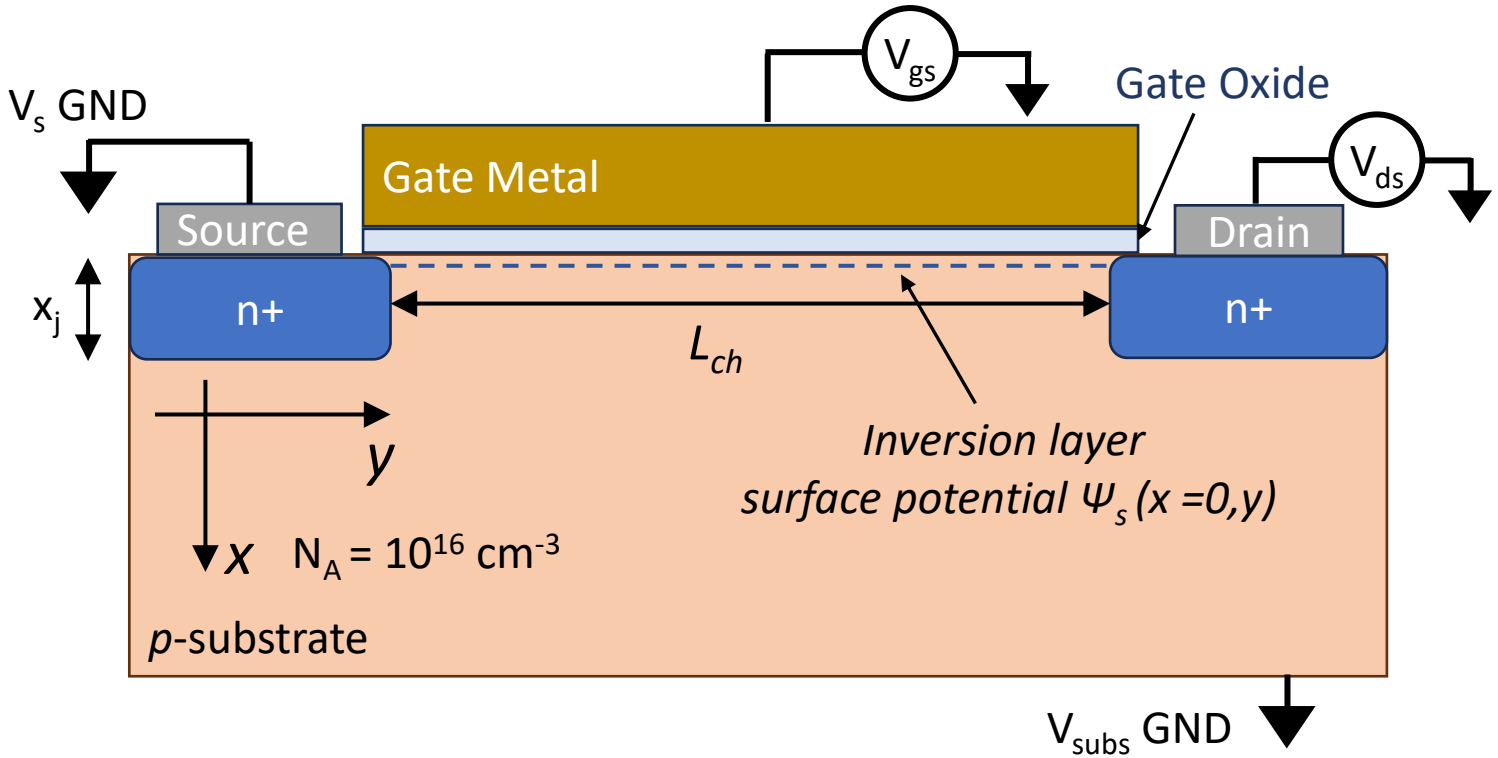
More technical:

Characteristic length 'λ'

'Electrostatic Integrity'

Start with long-channel MOSFET

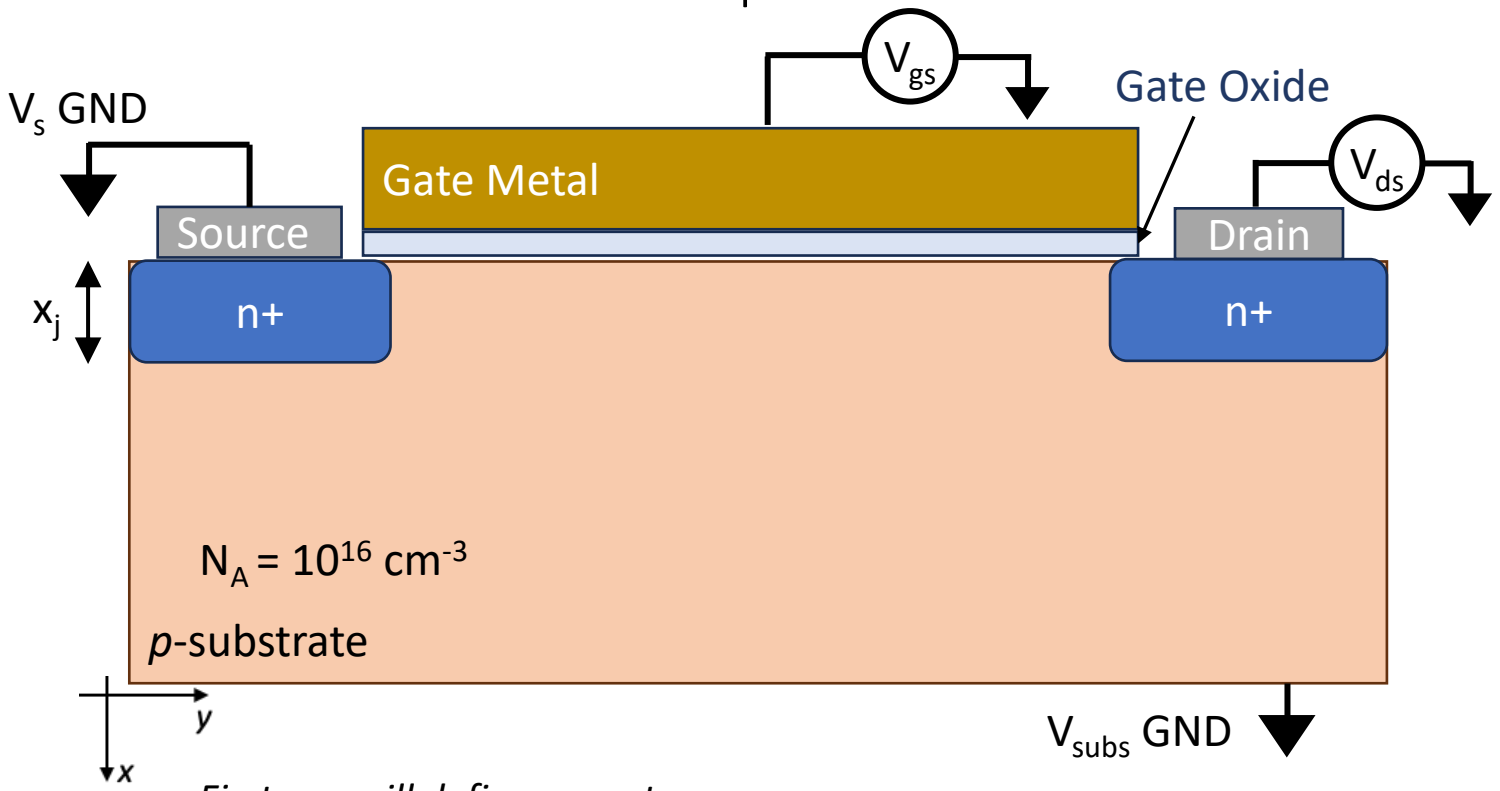
Definitions and Conventions



First, we will define some terms:

- x axis – longitudinal – direction of gate electric fields (consistent with MOSCAP)
- y axis – transverse – direction of flow of current
  Cutline x = 0, $\Psi_s(x = 0,y)$ – Inversion Layer
- L_{ch} – Channel length distance between n+ regions
- C_{ox} Gate Capacitance per unit area ε_{ox}/t_{ox}
- x_j – Junction depth
- Biasing : $V_{s,subs}$ -> GND, V_{gs} (x) and V_{ds} (y)

W_{dep} for 3 biasing conditions



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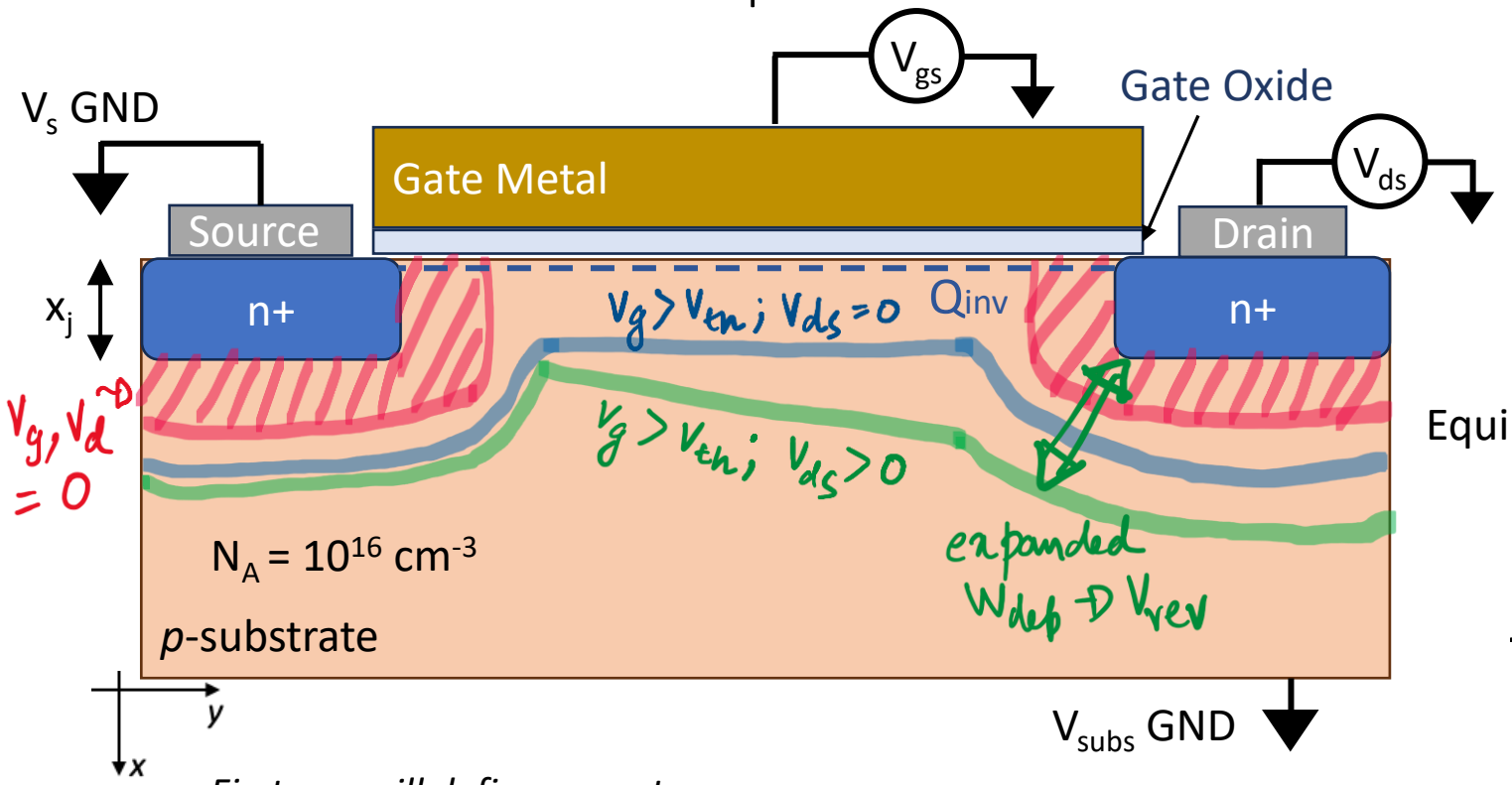
Now we will plot the Depletion Width (W_{dep}) for three biasing conditions
Assume $V_{fb} = 0$, ideal conditions

(a) $V_{gs} = 0$, $V_{ds} = 0$

(b) $V_{gs} > V_{th}$, $V_{ds} = 0$

(c) $V_{gs} > V_{th}$, $V_{ds} > 0$

W_{dep} for 3 biasing conditions



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- x axis – longitudinal – direction of gate electric fields (consistent with MOSCAP)
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Now we will plot the Depletion Width (W_{dep}) for three biasing conditions
Assume $V_{fb} = 0$, ideal conditions

Equilibrium

(a) $V_{gs} = 0$, $V_{ds} = 0$
Off condition, flatbands
No Inversion layer in channel

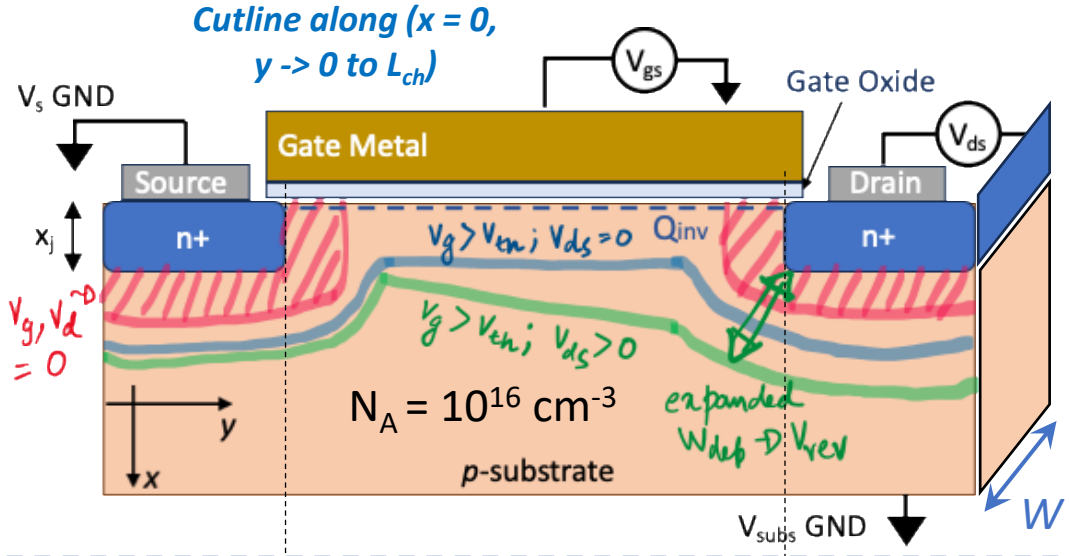
(b) $V_{gs} > V_{th}$, $V_{ds} = 0$
Inversion layer formed; W_{dep} grows as channel inverted and depleted

(c) $V_{gs} > V_{th}$, $V_{ds} > 0$
Inversion Layer + Current (E_y)
W_{dep} further grows V_{rev}
n+/p drain/subs junction - asymmetric

Non-Equilibrium

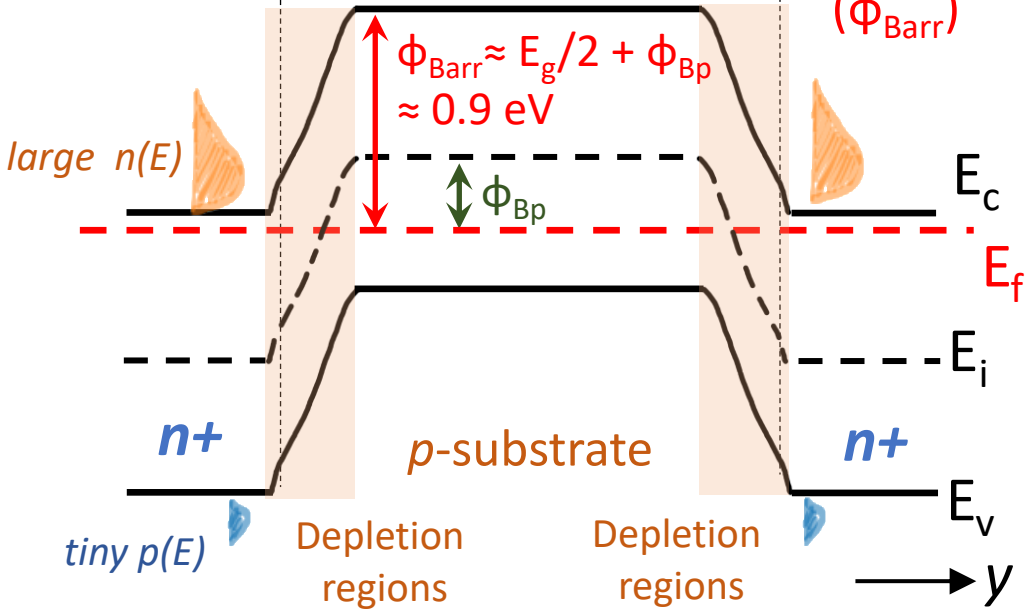
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Band Diagram (*x axis*) for 3 biasing conditions



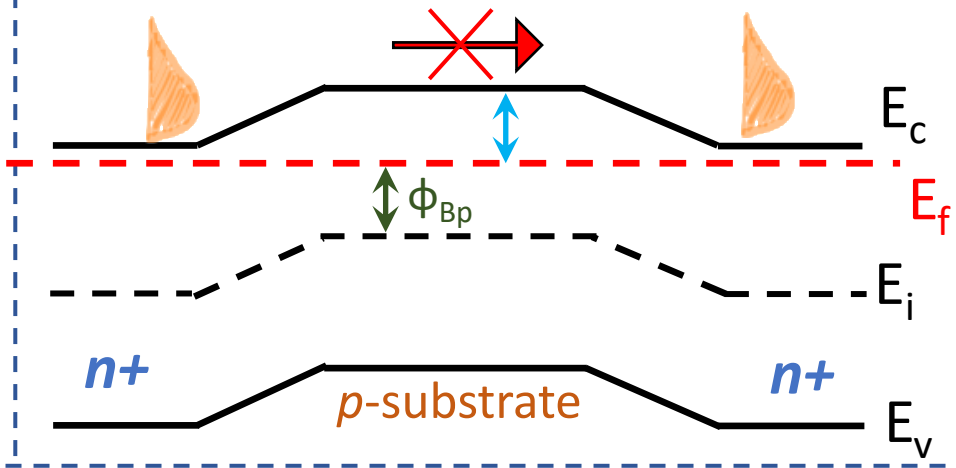
(a) $V_{gs} = 0$, $V_{ds} = 0$ {flatband}

OFF-state (ϕ_{Barr})



(b) $V_{gs} > V_{th}$, $V_{ds} = 0$ [Equilibrium]

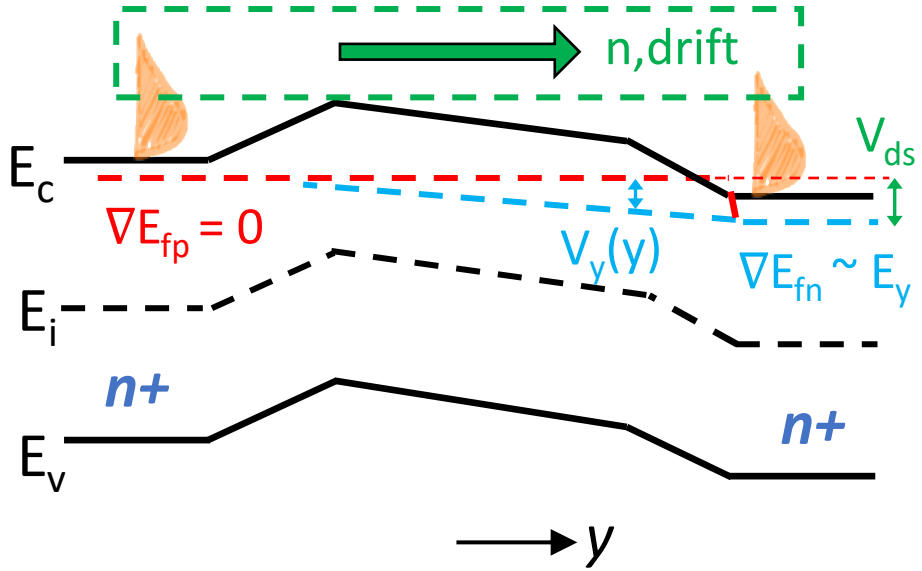
$\phi_{Barr} < E_g/2 - \phi_{Bp} \approx 0.2$ eV



Still No Current

No E_y/V_y
n(E) source & drain align.

(c) $V_{gs} > V_{th}$, $V_{ds} > 0$ [Non-Equilibrium]



n,drift →
$J_{n,drift}$ ←

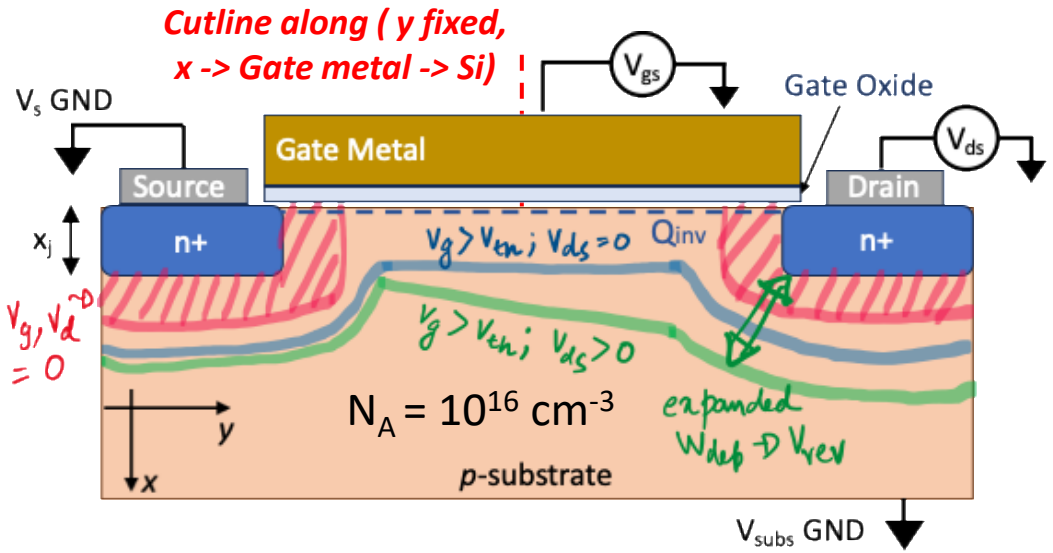
$$J_n = \frac{Q^*_{inv}}{t_{inv}} \mu_{eff} E_y$$

$$I_d = W t_{inv} \frac{Q_{inv}}{t_{inv}} \mu_{eff} E_y = W Q_{inv} \mu_{eff} E_y$$

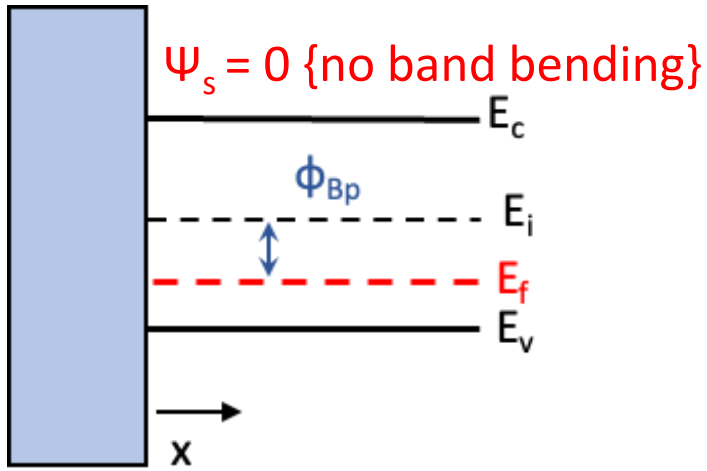
y = 0 to L_{ch} -> $V_y(y)$ = 0 to V_{ds}

* Q_{inv} in C/cm^{-2}

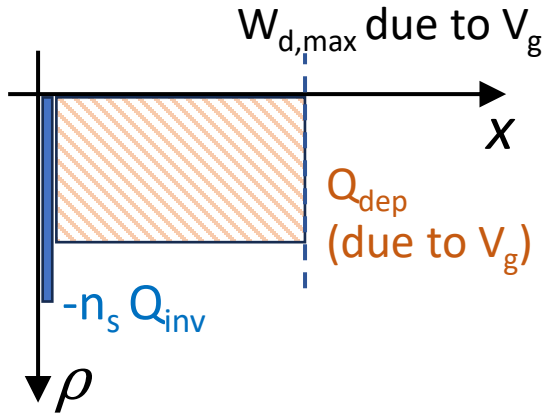
Band Diagram (*x axis*) for 3 biasing conditions

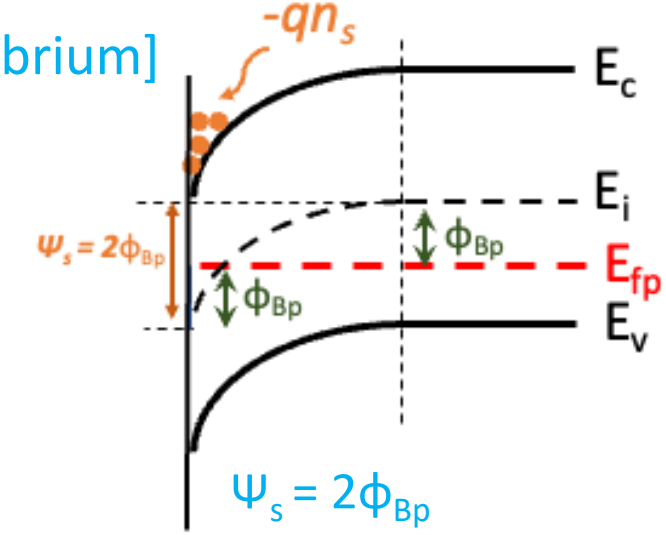


(a) $V_{gs} = 0$, $V_{ds} = 0$ {flatband}



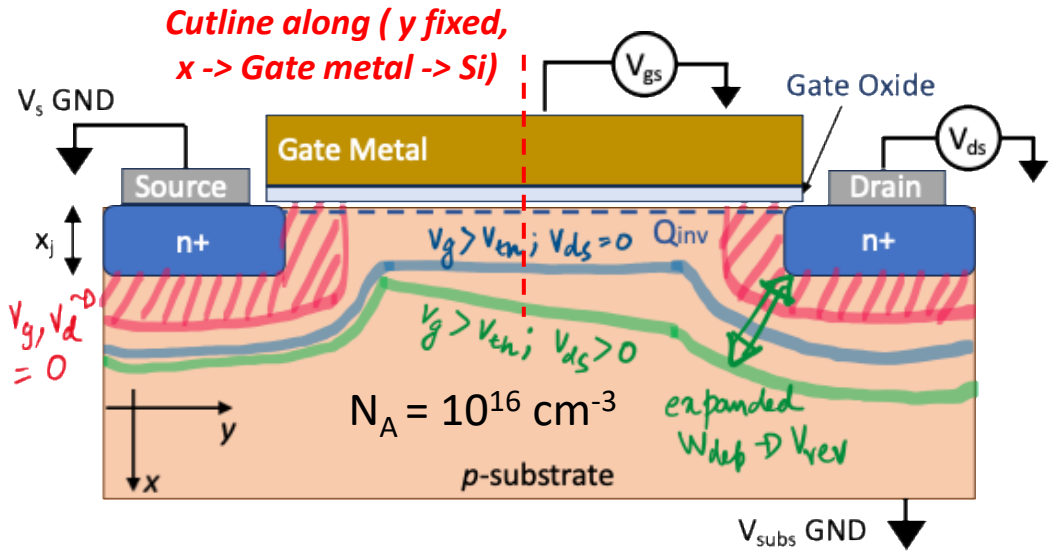
(b) $V_{gs} > V_{th}$, $V_{ds} = 0$ [Equilibrium]



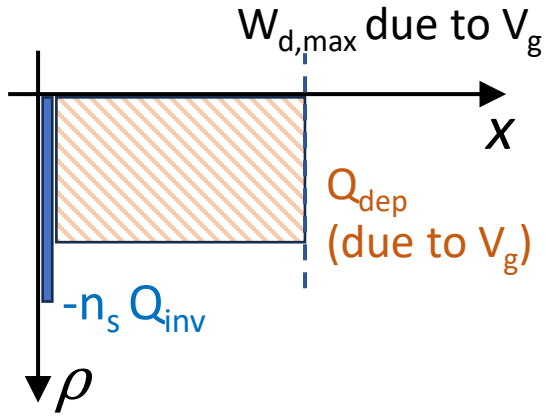


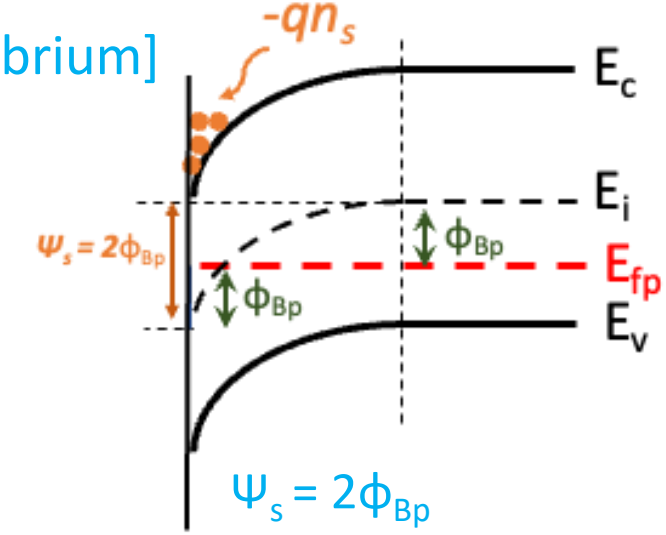
(c) $V_{gs} > V_{th}$, $V_{ds} > 0$ [Non-Equilibrium]

Band Diagram (*x axis*) for 3 biasing conditions

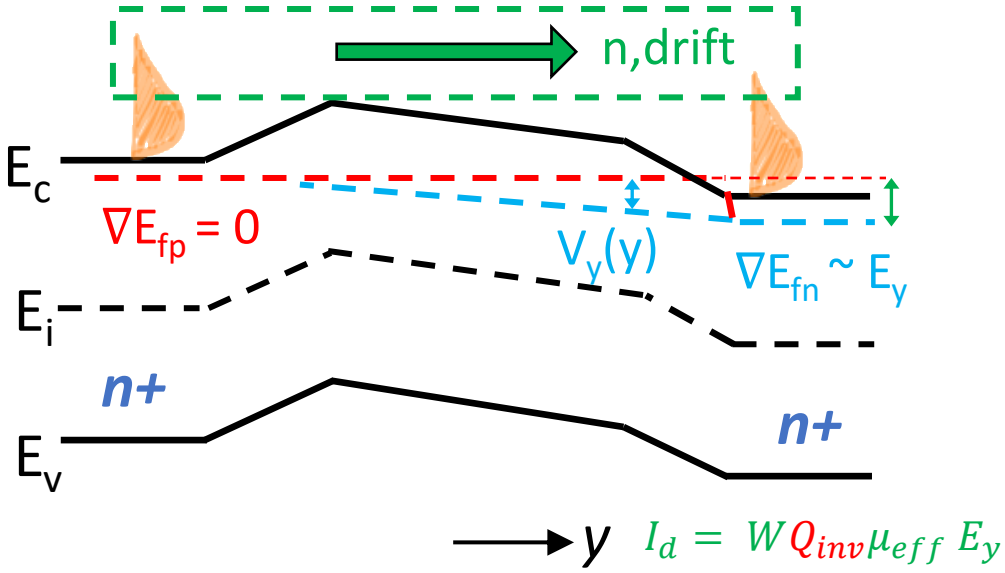


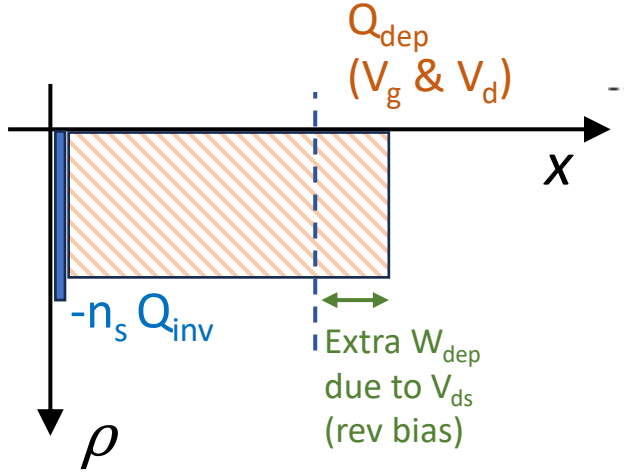
(b) $V_{gs} > V_{th}$, $V_{ds} = 0$ [Equilibrium]



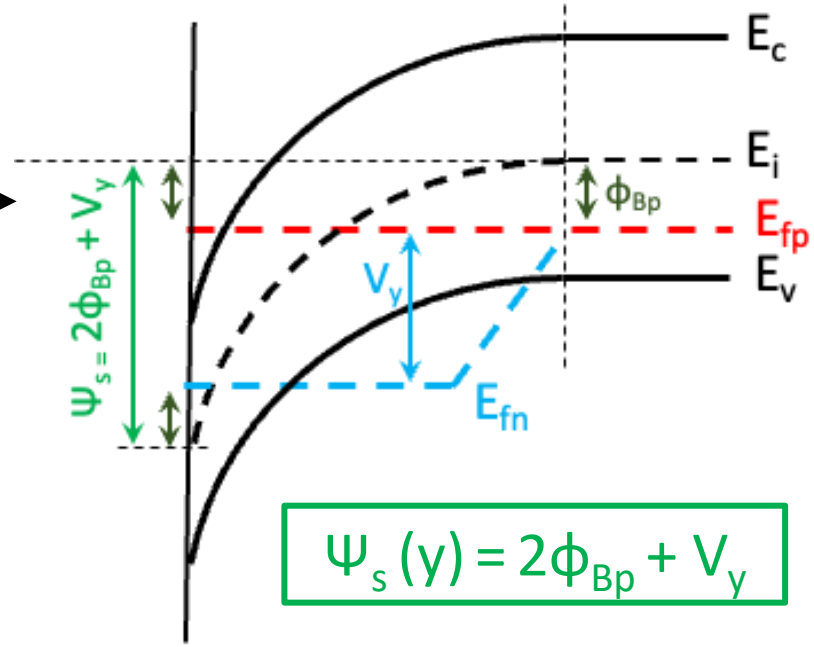


(c) $V_{gs} > V_{th}$, $V_{ds} > 0$ [Non-Equilibrium]



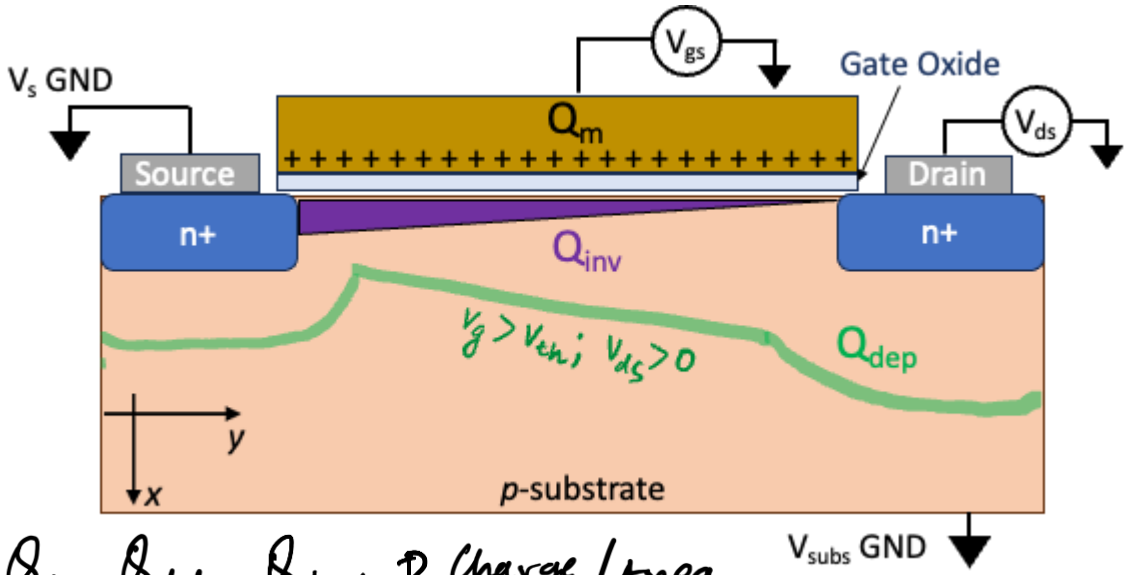


$Q_{dep}(y) \rightarrow \psi_s(y)$!

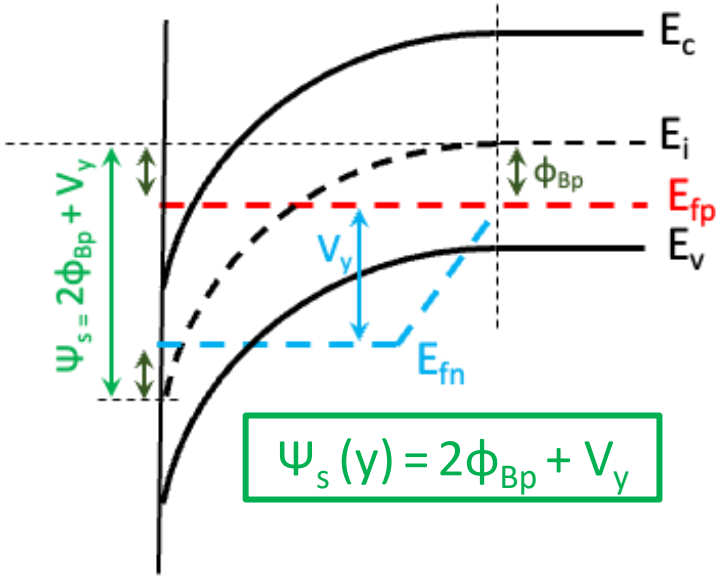


$$\psi_s(y) = 2\phi_{Bp} + V_y$$

Deriving $Q_{inv}(y)$ and I_{ds}, $V_{gs}>V_{th}$, $V_{ds}>0$



Q_m, Q_{dep}, Q_{inv} → Charge / Area



$$\Psi_s(y) = 2\phi_{Bp} + V_y$$

@ every point 'y' → fixed & metal equipotential

$$Q_m = Q_{dep}(y) + Q_{inv}(y) \quad (1)$$

Q_{dep}: increasing with y; $\psi_s(y)$; Q_{inv}: decreasing with y

Recall for MOSCAP: $V_g = V_{fb} + V_{ox} + V_{si}$

$$V_g = V_{fb} + \frac{Q_{dep} + Q_{inv}}{C_{ox}} + \psi_s \quad (2)$$

How does (2) transform for MOSFET?

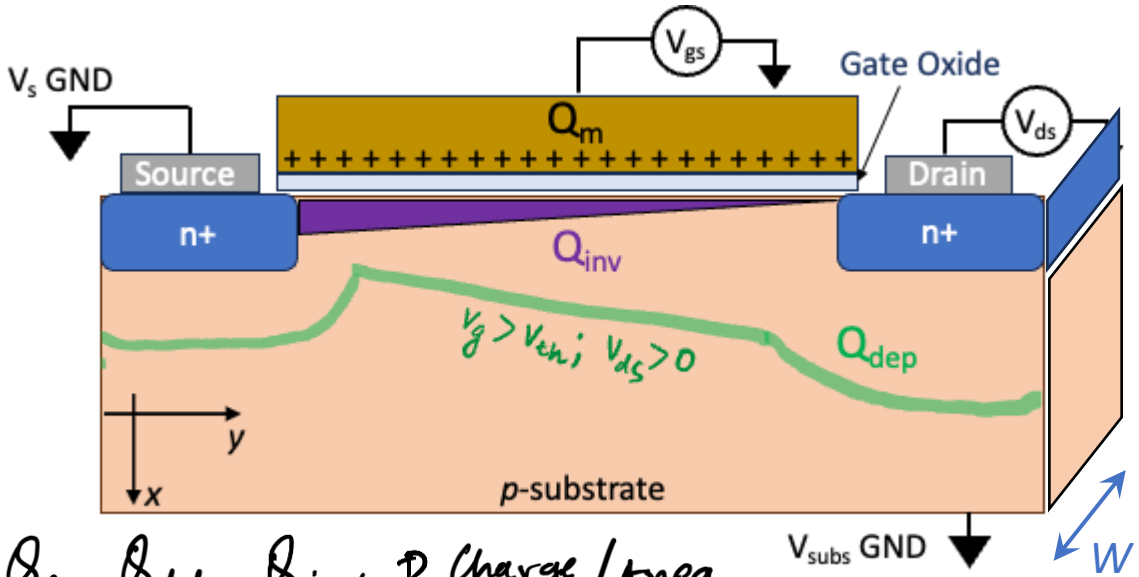
$$V_{gs} = V_{fb} + \frac{Q_{dep}(y) + Q_{inv}(y)}{C_{ox}} + \psi_s(y) \quad (3)$$

Because Q_{dep}, Q_{inv}, ψ_s change with y

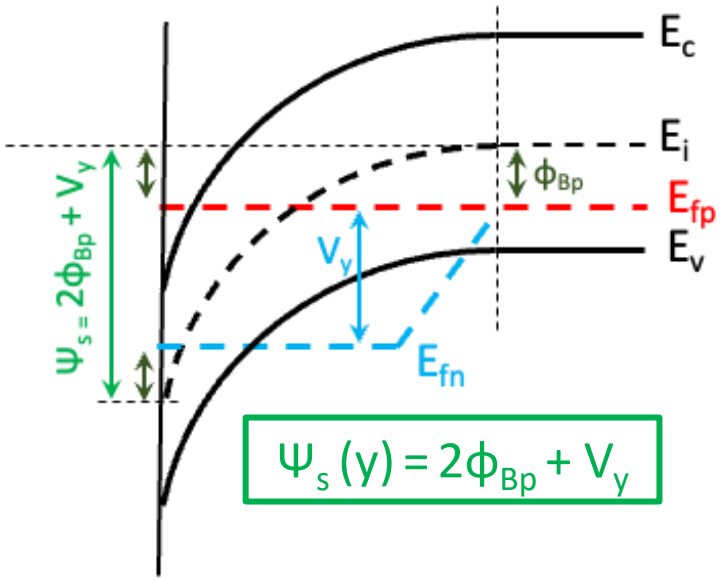
To calculate I_{ds} we need Q_{inv}

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{fb} - \psi_s(y) - \frac{Q_{dep}(y)}{C_{ox}}\right] \quad (4)$$

Deriving $Q_{inv}(y)$ and I_{ds}



Q_m, Q_{dep}, Q_{inv} → Charge / Area



$$\Psi_s(y) = 2\phi_{Bp} + V_y$$

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{fb} - \Psi_s(y) - \frac{Q_{dep}(y)}{C_{ox}}\right] \quad (4)$$

Subs:- $\Psi_s(y) = V(y) + 2\phi_{BP}$ & $Q_{dep}(y) = \sqrt{2q\epsilon_{si}N_A\Psi_s(y)}$

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{fb} - (2\phi_{Bp} + V_y) - \frac{1}{C_{ox}}\sqrt{2q\epsilon_{si}N_A(2\phi_{Bp} + V_y)}\right] \quad (5)$$

$I_{ds} = W\, Q_{inv}(y)\, \mu_{n,eff}\, \frac{dV_y}{dy}$ → Electric field along y due to V_{ds}.

W → Transistor width; $\mu_{n,eff}$ → effective mobility

$$\int_0^{L_{ch}} I_{ds}\, dy = W\, \mu_{n,eff} \int_0^{V_{ds}} Q_{inv}(y)\, dV_y \quad (6)$$

Integrating Q_{inv} directly leads to Ihantolla-Moll Model (Appendix -1)

(Not needed for our course – We will approximate Q_{inv})

Taylor Series & Simplified Q_{inv}

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{fb} - (2\phi_{BP} + V_y) - \frac{1}{C_{ox}}\sqrt{2q\epsilon_{si}N_A(2\phi_{BP}+V_y)}\right] \quad (5)$$

{ Taylor series for $V_y \ll \phi_{BP}$ } $\Rightarrow$ $(2\phi_{BP} + V_y)^{1/2} = \sqrt{2\phi_{BP}} + \frac{V_y}{2\sqrt{2\phi_{BP}}} + \dots$

IMP:- Valid for small V_{ds}

Ignoring $O>2$ terms and Substituting above :-

$$Q_{inv}(y) \approx C_{ox}\left[V_{gs} - V_{fb} - 2\phi_{BP} - \frac{\sqrt{2\epsilon_{si}qN_A(2\phi_{BP})}}{C_{ox}} - \left(1 + \frac{1}{C_{ox}} \times \frac{1}{2}\sqrt{\frac{2q\epsilon_{si}N_A}{2\phi_{BP}}}\right)V_y\right]$$

$\frac{\sqrt{2\epsilon_{si}qN_A(2\phi_{BP})}}{C_{ox}} = V_{tho}$

$\frac{1}{2}\sqrt{\frac{2q\epsilon_{si}N_A}{2\phi_{BP}}} = C_{dep0}$

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{tho} - \left(1 + \frac{C_{dep0}}{C_{ox}}\right)V_y\right]$$

$m \rightarrow$ body factor

Simplified $Q_{inv}(y)$

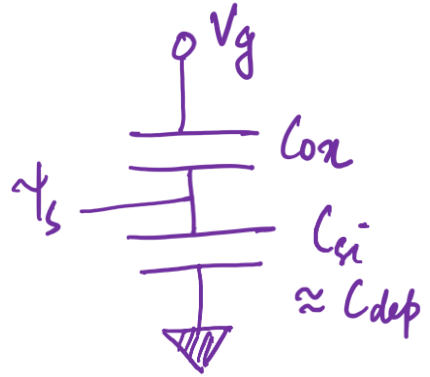
$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{tho} - mV_y\right] \sim (7)$$

$$Q_{dep} = qN_A W_{dep} = \sqrt{2qN_A\epsilon_{si}\psi_s}$$

$$\frac{dQ_{dep}}{d\psi_s} = C_{dep} = \frac{1}{2}\sqrt{\frac{2qN_A\epsilon_{si}}{\psi_s}}$$

$C_{dep0} \Rightarrow @\ V_{ds} = 0;\ @\ \psi_s = 2\phi_{BP}$

$$C_{dep0} = \frac{1}{2}\sqrt{\frac{2qN_A\epsilon_{si}}{2\phi_{BP}}}$$



$$m = \frac{dV_{gs}}{d\psi_s} = \left(1 + \frac{C_d}{C_{ox}}\right)$$

Physical Insights on $Q_{inv}(y)$

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{fb} - (2\phi_{BP} + V_y) - \frac{1}{C_{ox}}\sqrt{2q\epsilon_{Si}N_A(2\phi_{BP}+V_y)}\right]$$

{ Taylor series for $V_y \ll \phi_{BP}$ } $\Rightarrow (2\phi_{BP} + V_y)^{1/2} = \sqrt{2\phi_{BP}} + \frac{V_y}{2\sqrt{2\phi_{BP}}} + \dots$

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Ignoring $O>2$ terms and Substituting above :-

$$Q_{inv}(y) \approx C_{ox}\left[V_{gs} - V_{fb} - 2\phi_{BP} - \frac{\sqrt{2\epsilon_{Si}qN_A(2\phi_{BP})}}{C_{ox}} - \left(1 + \frac{1}{C_{ox}} \times \frac{1}{2}\sqrt{\frac{2q\epsilon_{Si}N_A}{2\phi_{BP}}}\right)V_y\right]$$

V_{tho} ; C_{dep_o}

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{tho} - \left(1 + \frac{C_{depo}}{C_{ox}}\right)V_y\right]$$

$m \rightarrow$ body factor

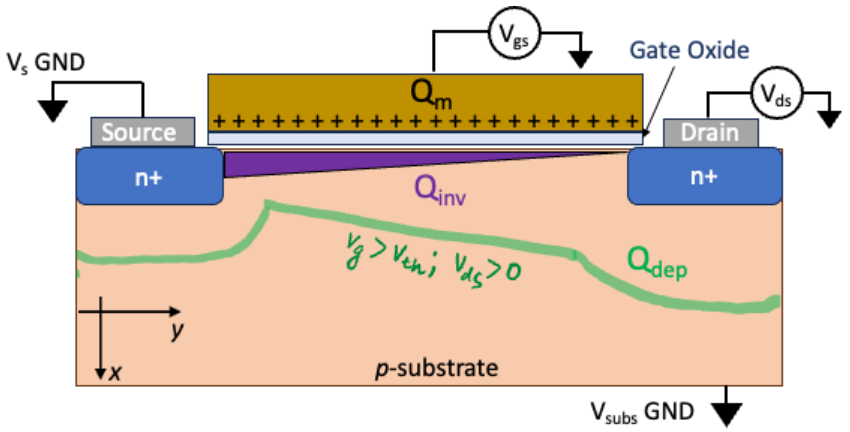
Simplified $Q_{inv}(y)$

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{tho} - mV_y\right] \sim (7)$$

Key Physical Insight

$Q_{inv}(y)$ depletes linearly from y = 0 (Source) -> y = L_{ch} (Drain)

$Q_{inv,\,Source}(0) = C_{ox}(V_{gs}-V_{th})$ ->
$Q_{inv,\,Drain}(L_{ch}) = C_{ox}(V_{gs}-V_{th}-mV_{ds})$



Since $Q_m = Q_{inv}(y) + Q_{dep}(y)$
$Q_{dep}(y)$ increases linearly with *y*
(First approximation)

Derivation of I_{ds} & Output Char (I_{ds} vs V_{ds})

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{tno} - m V_y\right] \quad (7)$$

$$I_{ds} = W \mu_{n,eff} Q_{inv}(y) \frac{dV_y}{dy} \quad (6)$$

$$I_{ds}\, dy = W \mu_{n,eff} Q_{inv}(y)\, dV_y$$

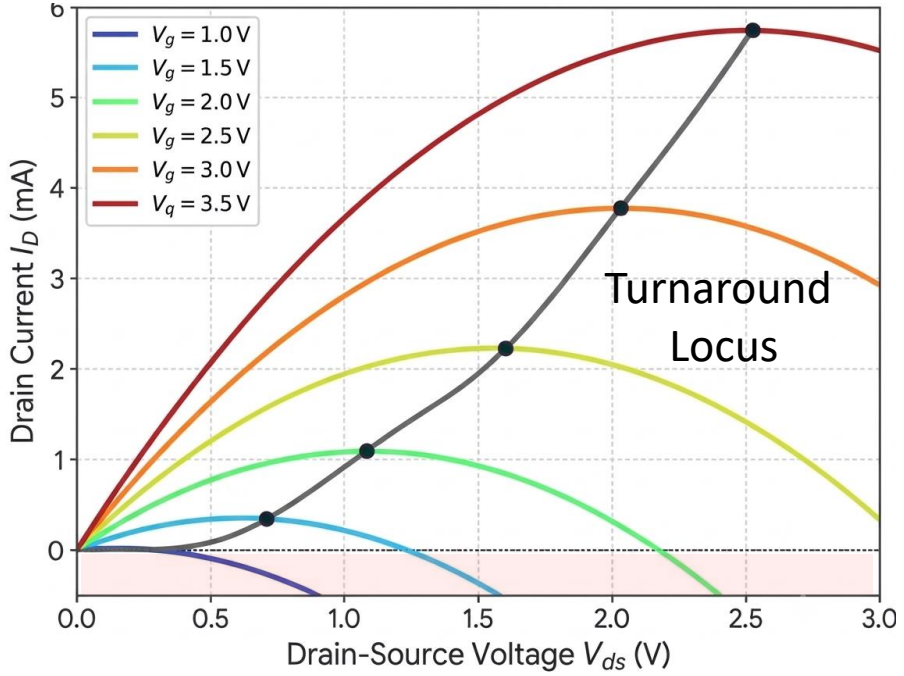
$$I_{ds} \int_0^{L_{ch}} dy = W \mu_{n,eff} \int_0^{V_{ds}} C_{ox}\left[V_{gs} - V_{tno} - m V_y\right] dV_y$$

$$I_{ds} = \frac{W}{L_{ch}} \mu_{n,eff} C_{ox} \left\{(V_{gs} - V_{tno}) V_{ds} - \frac{m V_{ds}^2}{2}\right\} \quad (8)$$

Let's Plot the model Predictions

Parameters taken by me: W/L =10 µm/1 µm, $\mu_{n,eff}$ = 500 $cm^2V^{-1}s^{-1}$,
C_{ox} =0.35 $\mu F/cm^2$ (t_{ox} = 10 nm), m = 1.2

(A) Output Characteristics (V_{gs} fixed, V_{ds} swept)



Turnaround Locus: I_{ds} hits maxima!

$$\frac{dI_{ds}}{dV_{ds}} = 0 \Rightarrow V_{ds,sat} = \frac{V_{gs} - V_{tn}}{m}$$

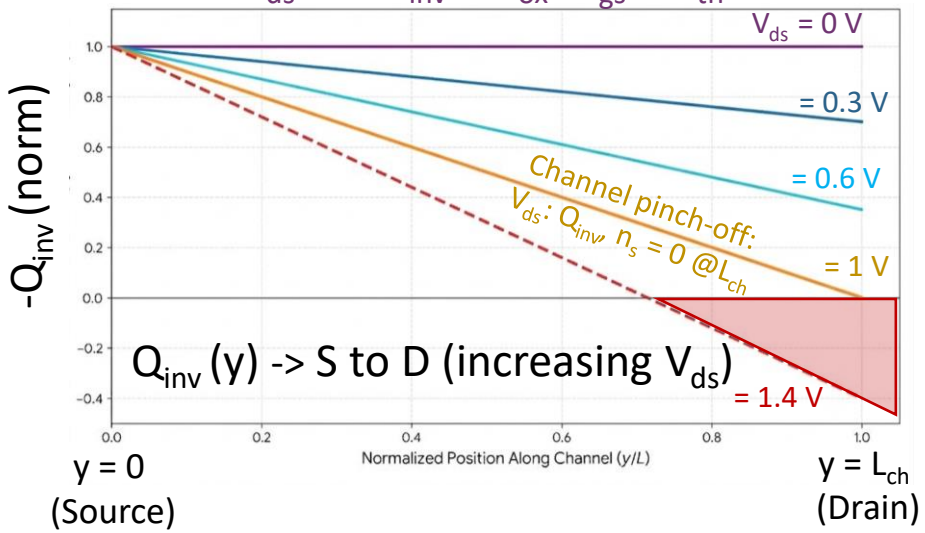
Turnaround (NDR) and $I_{ds} < 0$:
Incorrect Predictions of the model[Why?]

Look at Eqn 7: $Q_{inv}(y) = C_{ox}(V_{gs} - V_{th} - mV_y)$

Saturation in I_{ds} and $V_{ds,sat}$

Let us examine $Q_{inv}(y) = C_{ox}(V_{gs} - V_{th} - mV_y)$

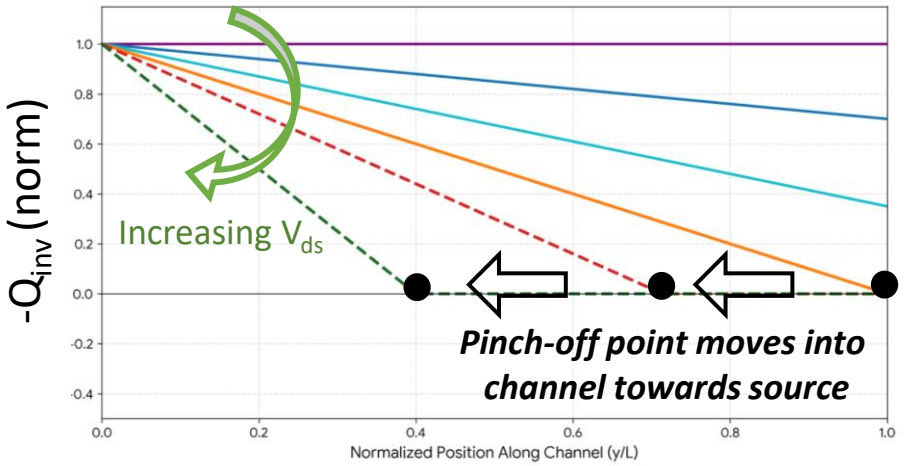
$V_{ds} = 0$ $Q_{inv} = C_{ox}(V_{gs} - V_{th})$



At $V_{ds} = (V_{gs}-V_{th})/m$
$Q_{inv}(L_{ch}) = 0$

For $V_{ds} > (V_{gs}-V_{th})/m$
-ve $-Q_{inv}$ {Unphysical!}
-ve area -> $I_{ds} = \int Q_{inv}$ reduces -> Turnaround

What actually happens V_{ds} > pinch-off?



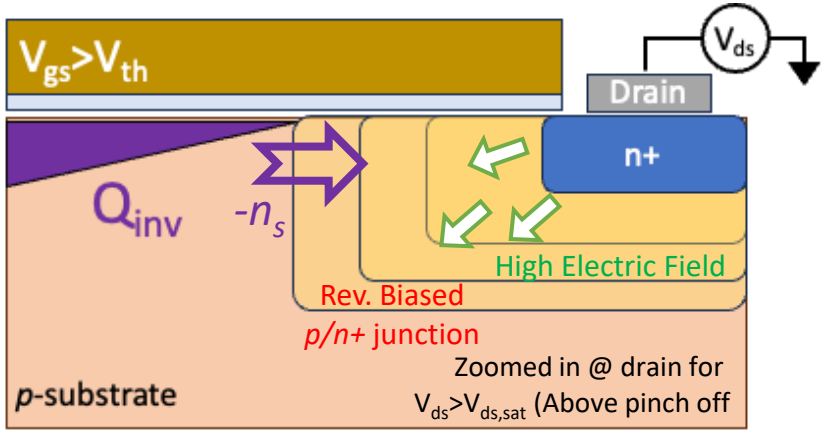
At $V_{ds} = (V_{gs}-V_{th})/m$
$Q_{inv}(L_{ch}) = 0$
y= L_{ch} -> Pinch off point

For $V_{ds} > (V_{gs}-V_{th})/m$
Pinch of point move inwards

Beyond Pinch off – model breaks down (no Q_{inv} for all *y*)

What happens to Ids V_{ds} > Pinch Off?

Common sense: I_{ds}=0 (no channel) or I_{ds} reduction $\int Q_{inv}$ reduces
– Not correct - Why?
I_{ds} NOT linear resistor [nqμE]



After pinchoff, ΔV_{ds} drops on Rev. biased p/n+ -> Large E_{field}
n_s swept -> Drain
I_{ds} saturates at $V_{ds,sat}$!
<J_{rev} indep. V_{rev}>

Corrected Output Char- Piecewise Clamp Model

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{tno} - m V_y\right] \quad (7)$$

$$I_{ds} = W \mu_{n,eff} Q_{inv}(y) \frac{dV_y}{dy} \quad (6)$$

$$I_{ds}\, dy = W \mu_{n,eff} Q_{inv}(y)\, dV_y$$

$$I_{ds}\int_0^{L_{ch}} dy = W \mu_{n,eff} \int_0^{V_{ds}} C_{ox}\left[V_{gs} - V_{tno} - m V_y\right] dV_y$$

$$I_{ds} = \frac{W}{L_{ch}} \mu_{n,eff} C_{ox}\left\{(V_{gs} - V_{tno})V_{ds} - \frac{m V_{ds}^2}{2}\right\} \quad (8)$$

Above expression valid for $Q_{inv} > 0$
or $V_{ds} < V_{ds_sat} = (V_{gs} - V_{th})/m$
For $V_{ds} > V_{ds_sat}$ -> I_{ds} saturates

(A) For linear regime; $V_{ds} \le V_{ds,sat}$

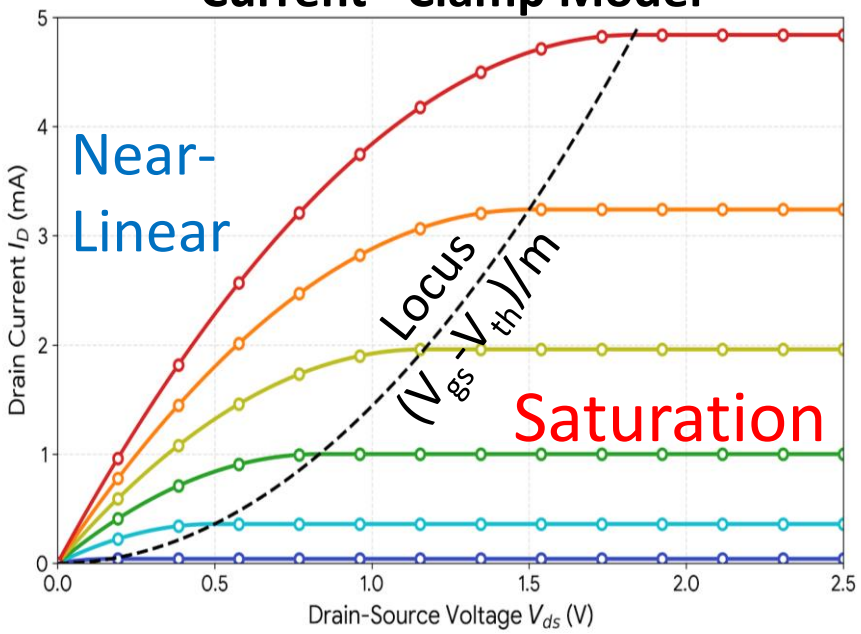
$$I_{ds} = \frac{W}{L_{ch}} \mu_{n,eff} C_{ox}\left[(V_{gs} - V_{tn})V_{ds} - \frac{m V_{ds}^2}{2}\right]$$

(B) For Saturation; $V_{ds} > V_{ds,sat}$

Substituting $V_{ds,sat} = (V_{gs} - V_{th})/m$

$$I_{ds} = \frac{W}{L_{ch}} \mu_{n,eff} C_{ox}\left[\frac{(V_{gs} - V_{tn})^2}{2}\right]$$

Current –Clamp Model



I_{ds} vs V_{gs} (fixed V_{ds}) – Transfer Characteristics

Transfer Characteristics I_{ds} vs V_{gs} (fixed V_{ds}) measured to extract V_{th}, $\mu_{n,eff}$: Small V_{ds}

$$I_{ds} = \frac{W}{L_{ch}} \mu_{n,eff} C_{ox} \left[(V_{gs} - V_{th}) V_{ds} - \frac{m V_{ds}^2}{2} \right]$$

$$\text{Slope} = \frac{\partial I_{ds}}{\partial V_{gs}} = \frac{W}{L_{ch}} \boldsymbol{\mu_{n,eff}} C_{ox} V_{ds}$$

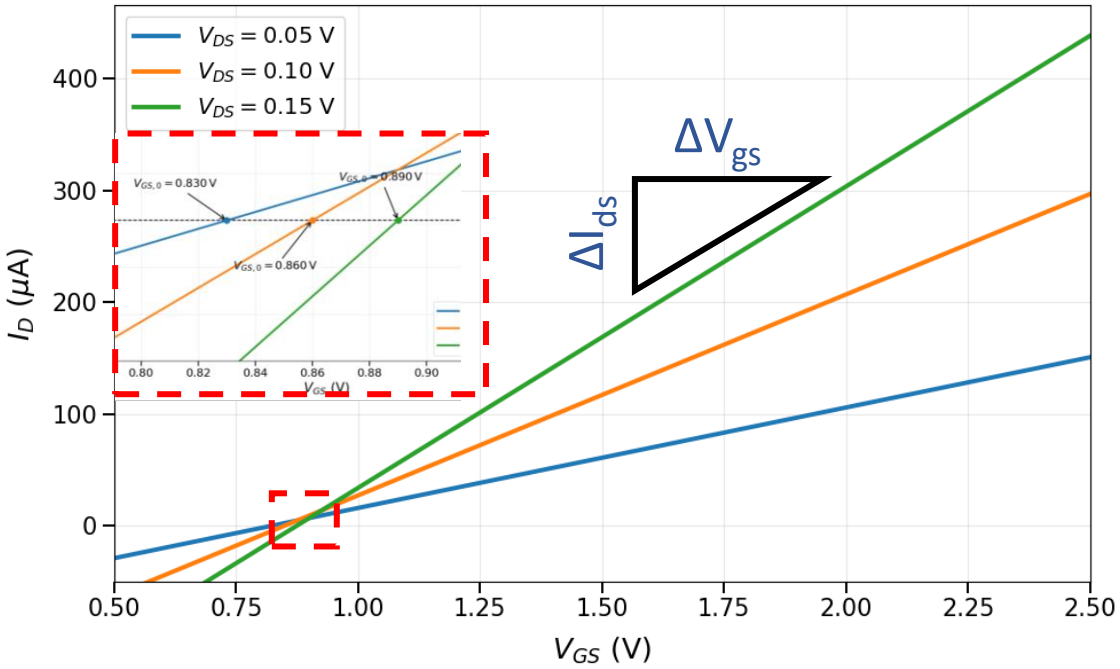
$$V_{gs}\ \text{Intercept} = \boldsymbol{V_{th}} + \frac{m V_{ds}}{2}$$

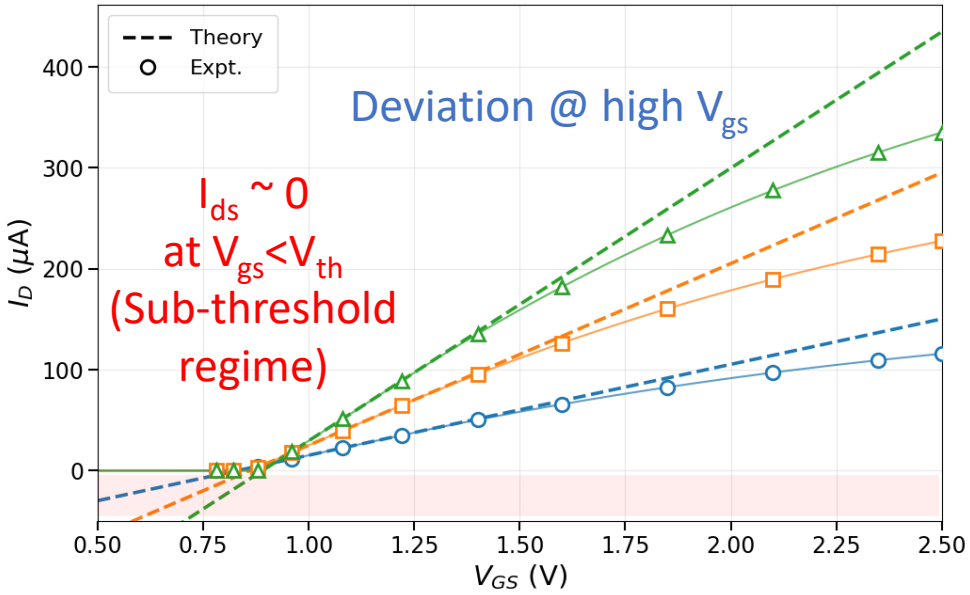
Small V_{ds} Can be ignored

However, deviations in measured I_{ds} vs V_{gs} vs Theory makes extraction of parameters non-trivial

(A) Not linear - significant deviation at high V_{gs}

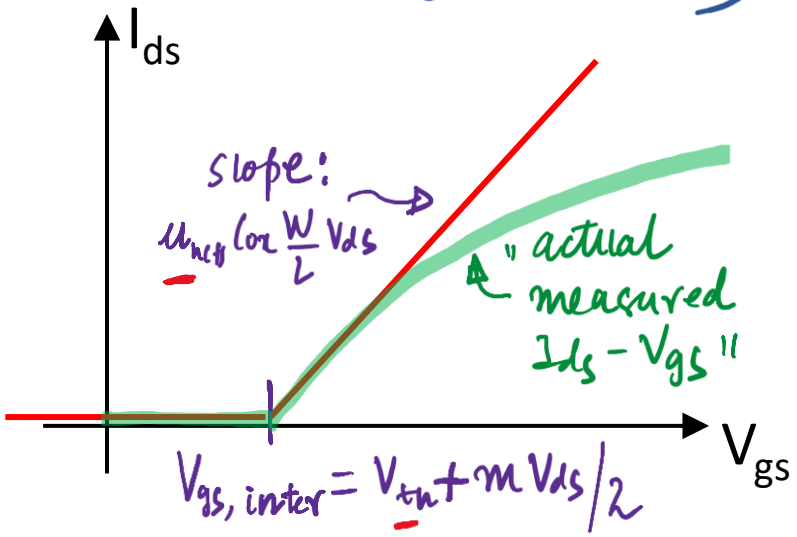
(B) I_{ds} small but positive in sub-threshold regime ($V_{gs} < V_{th}$)



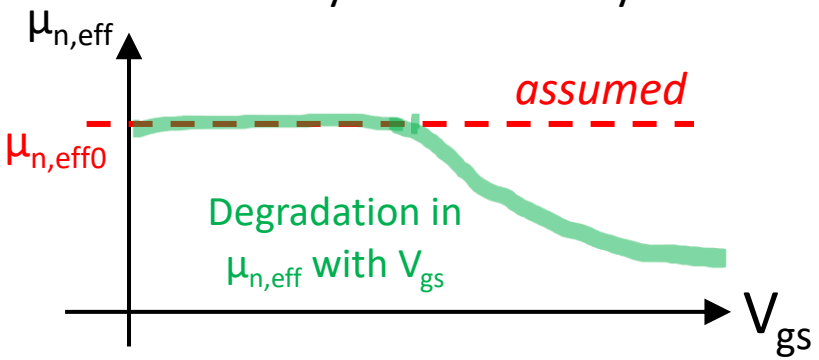


Transfer Characteristics and Mobility Degradation

Small V_{ds} $I_{ds} = K'\left[(V_{gs}-V_{th}) - \frac{mV_{ds}^2}{2}\right]$

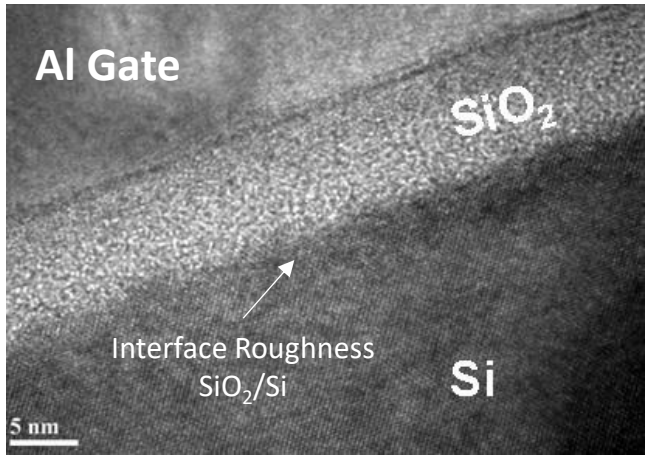


Why degradation in current with V_{gs}?
A: We assume $\mu_{n,eff}$ to be constant
Hardly the case! Why?

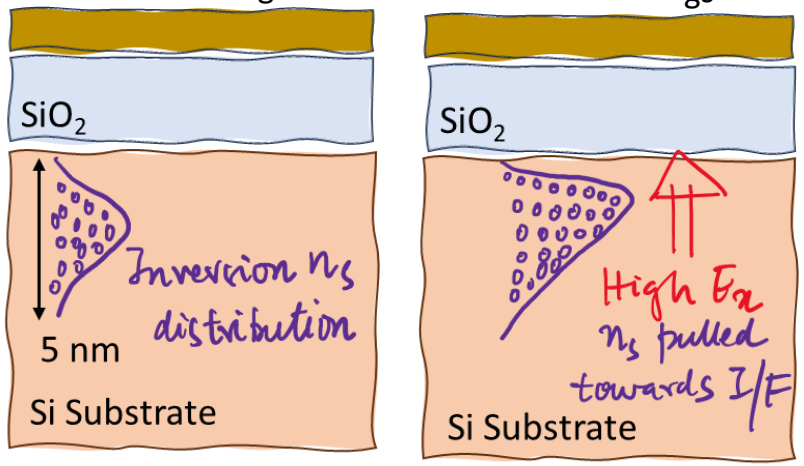


Cross-sectional TEM

Si/SiO_2 interface far from being smooth!

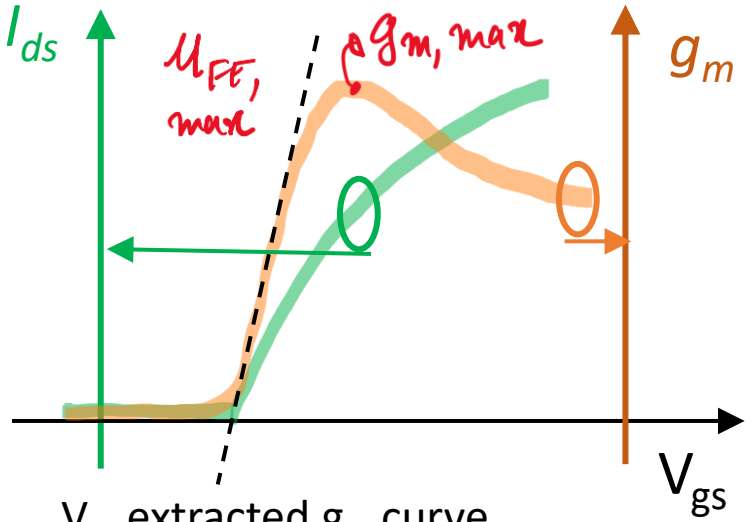


Low - V_{gs} High - V_{gs}



At Higher V_{gs} centroid shifts to I/F -> higher surface-roughness scattering

Extraction of V_{th}, μ for MOSFET



V_{th} extracted g_m curve

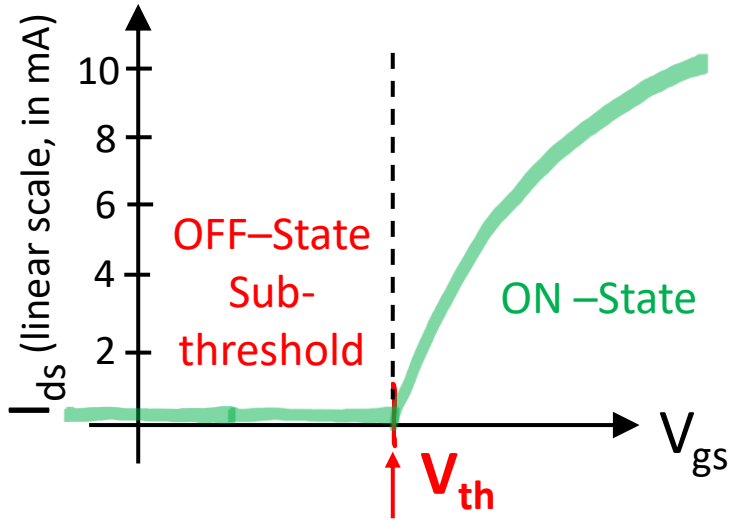
$g_m(V_{gs}) = \frac{\partial I_{ds}}{\partial V_{gs}}\Big|_{\text{fixed } V_{ds}\{\text{small}\}}$

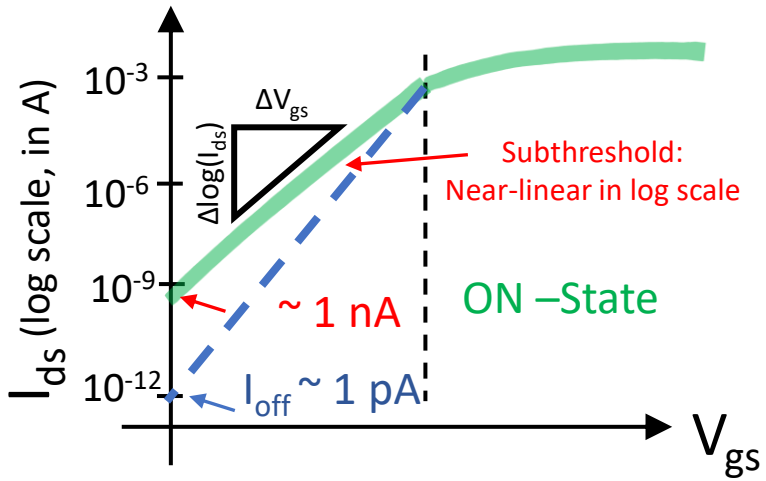
$\mu_{FE}(V_{gs}) = g_m \Big/ \left(\frac{W}{L_{ch}} C_{ox} V_{ds}\right)$

g_m (Transconductance) &
μ_{FE} (Field Effect Mobility)
Key figure of merit for transistor performance.

Subthreshold Conduction and Swing

Till Now: I_{ds} (V_{gs}>V_{th})
What happens I_{ds} (V_{gs}< V_{th}),
Q_{inv} = 0

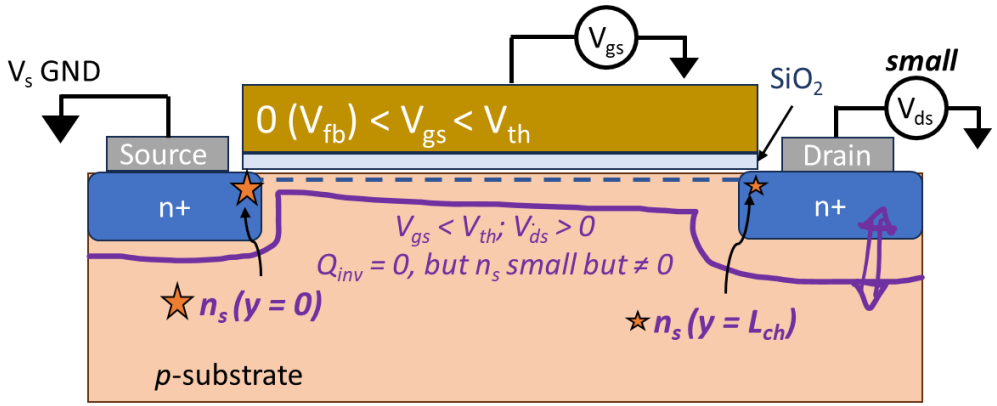


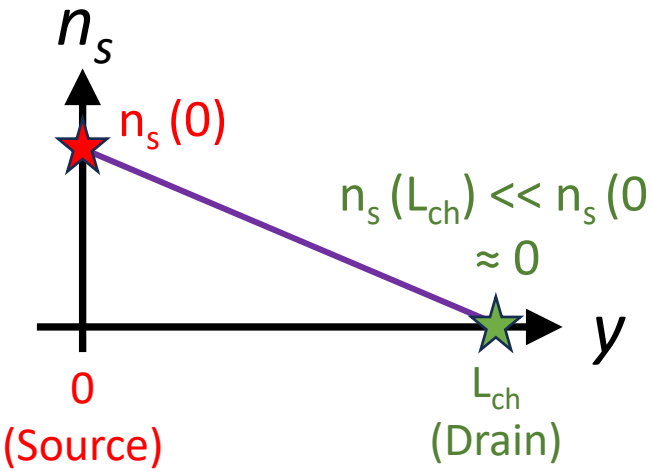


Subthreshold Swing, $SS = \frac{dV_{gs}}{d(log_{10} I_{ds})}$, in mV/dec

How much ΔV_{gs} needed to ΔI_{ds} by 10x(decade)
SS – Critical Parameter – Why? Steeper SS -> lower I_{off}
I_{off} - Leakage Power billions of MOSFETs in modern ICs

} Nature of SS conduction?





Since $n_s(y)$ very small $I_{n,drift} \approx 0$
But! Notice $\nabla n_s(y)$ due to V_{ds} (V_y): V_{rev}

I_{ds} (V_{gs}< V_{th}) due to Diffusion of n_s S->D!

$$I_{ds}^{n,diff} = -qAD_n \nabla n_s(y)$$

$$I_{ds}^{n,diff} \approx -qAD_n \frac{\Delta n_s}{\Delta y}$$

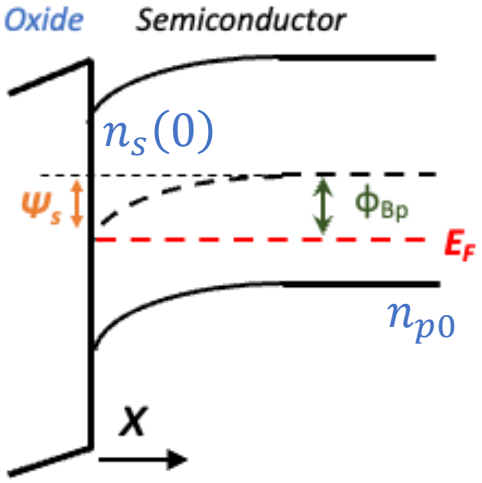
Simlified $\nabla n_s(y)$
Full Derivation Appendix II

$$I_{ds}^{SS} \approx -qAD_n \frac{n_s(0) - n_s(L_{ch})}{0 - L_{ch}}$$

$$I_{ds}^{SS} \approx qAD_n \frac{n_s(0)}{L_{ch}}$$

Derivation of Subthreshold Swing

At Source equ. MOSCAP



$$I_{ds}^{SS} \approx qAD_n \frac{n_s(0)}{L_{ch}}$$

$$n_s(0) = n_{p0}\exp(\psi_s/V_t)$$

$$I_{ds}^{SS} = \frac{qAD_n n_{p0}}{L_{ch}} \exp(\psi_s/V_t)$$

$$\ln(I_{ds}^{SS}) = \ln(I_0) + \frac{\psi_s}{V_t}$$

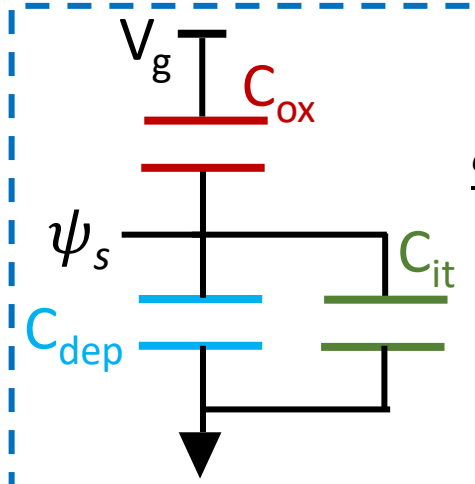
$$\frac{d(\log_{10} I_{ds}^{SS})}{d\psi_s} = \frac{1}{V_t \ln(10)}$$

$$SS = \frac{dV_{gs}}{d(\log_{10} I_{ds})} = \frac{dV_{gs}}{d\psi_s} \frac{d\psi_s}{d(\log_{10} I_{ds})}$$

$$SS = V_t \ln(10)\,[1 + (C_{dep} + C_{it})/C_{ox}]$$

$$SS = 60\ [1 + (C_{dep} + C_{it})/C_{ox}]\ mV/dec$$

$SS > 60$ mV/decade – What is the implication?

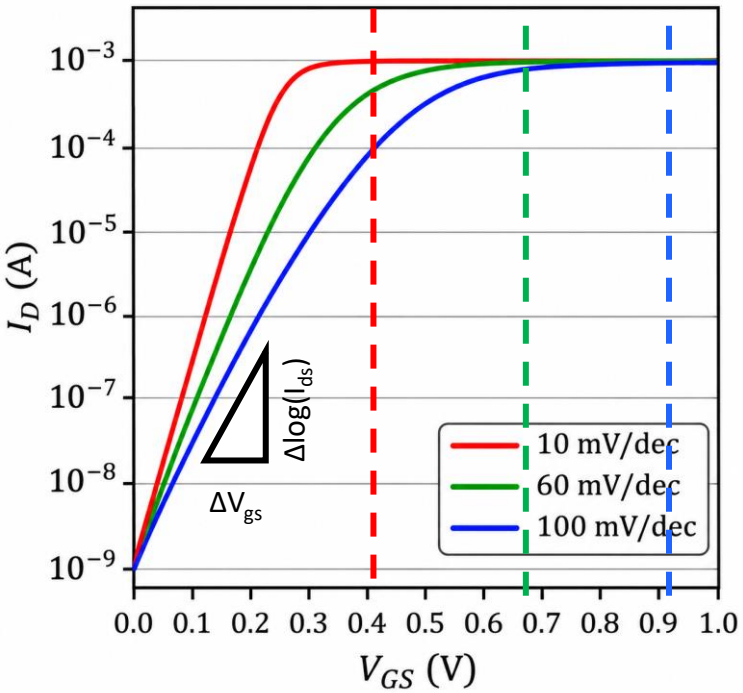


Body Factor '*m*'

$$\frac{dV_{gs}}{d\psi_s} = 1 + \frac{C_{dep}}{C_{ox}} \text{ (ideal)}$$

$$= 1 + \frac{C_{dep}+C_{it}}{C_{ox}}$$

(with C_{it})



(A) Ideal : 60 mV/dec
6 decades I_{on}/I_{off} ~ 360 mV
+ V_{ov} Margins = 300 mV
V_{dd} = 0.66 V

(B) Practical ~100 mV/dec
6 decades I_{on}/I_{off} ~ 600 mV
$V_{dd} \approx$ 0.9 -1 V
– V_{dd} scaling big challenge

(C) Subthermionic Transistors
SS<<60 mV/dec V_{dd} ~350 mV

Appendix– 1: Ihantola–Moll Model

$$I_{ds} L_{ch} = W \mu_{n,eff} C_{ox} \int_0^{V_{ds}} \left[\underbrace{V_{gs} - V_{fb} - 2\phi_{Bf}}_{①} - \underbrace{V_y}_{②} - \underbrace{\frac{\sqrt{2q\epsilon_{Si} N_A (2\phi_{BP} + V_y)}}{C_{ox}}}_{③} \right] dV_y$$

$$① \int_0^{V_{ds}} (V_{gs} - V_{fb} - 2\phi_{Bp})\, dV_y = (V_{gs} - V_{fb} - 2\phi_{Bp}) V_{ds} \qquad ② \int_0^{V_{ds}} V_y\, dV_y = \frac{1}{2} V_{ds}^2$$

$$③ \int_0^{V_{ds}} \frac{\sqrt{2\epsilon_{Si} q N_A}}{C_{ox}} (2\phi_{Bp} + V_y)^{1/2}\, dV_y = \frac{2}{3} \frac{\sqrt{2q\epsilon_{Si} N_A}}{C_{ox}} \left\{ (2\phi_{Bp} + V_{ds})^{3/2} - (2\phi_{Bp})^{3/2} \right\}$$

$$I_{ds} = \frac{W}{L} \mu_{n,eff} C_{ox} \left[\left\{ V_{gs} - V_{fb} - 2\phi_{Bp} - \frac{V_{ds}}{2} \right\} V_{ds} - \frac{2}{3} \frac{\sqrt{2q\epsilon_{Si} N_A}}{C_{ox}} \left\{ (2\phi_{Bp} + V_{ds})^{3/2} - (2\phi_{Bp})^{3/2} \right\} \right] ⑦$$

"IHANTOLA - MOLL MODEL"

Let's Plot the transfer and output characteristics with the Ihantola–Moll Model (Python code: DIY)

Parameters taken by me: W/L =10 µm/1 µm, $\mu_{n,eff}$ = 500 $cm^2V^{-1}s^{-1}$,
C_{ox} =0.35 $\mu F/cm^2$ (t_{ox} = 10 nm), N_A =$10^{16} cm^{-3}$ -> ϕ_{Bp} =0.35 V

Simplified Expression for I_{ds} – Low V_{ds}

$$I_{ds} = \frac{W}{L}\mu_n C_{ox}\left[\left\{V_{gs} - V_{fb} - 2\phi_{Bp} - \frac{V_{ds}}{2}\right\}V_{ds} - \frac{2}{3}\frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}}\left\{(2\phi_{Bp}+V_{ds})^{3/2} - (2\phi_{Bp})^{3/2}\right\}\right] \quad (7)$$

To simplify :- Taylor series expansion $(2\phi_{Bp}+V_{ds})^{3/2}$ for small $V_{ds} \ll 2\phi_{Bp}$

$$(2\phi_{Bp}+V_{ds})^{3/2} - (2\phi_{Bp})^{3/2} = V_{ds}\frac{3}{2}\sqrt{2\phi_{Bp}} + \frac{V_{ds}^2}{4}\frac{3}{2}(2\phi_{Bp})^{-1/2} - \frac{1}{16}V_{ds}^3(2\phi_{Bp})^{-3/2} + \cdots$$

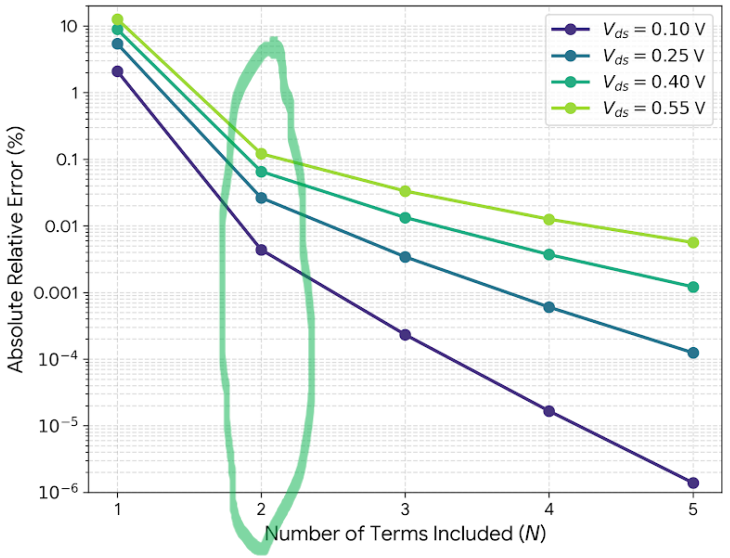
{Q: How many terms to consider?}

2 terms only.

$$I_{ds} = \frac{W}{L_m}\mu_{n,eff}C_{ox}\left[\left(V_{gs} - V_{fb} - 2\phi_{Bp} - \frac{Q_{dep0}}{C_{ox}}\right)V_{ds} - \frac{1}{2}\left(1+\frac{C_{dep}}{C_{ox}}\right)V_{ds}^2\right]$$

V_{tn}

NUMERICALLY ⇒ Error vs term.

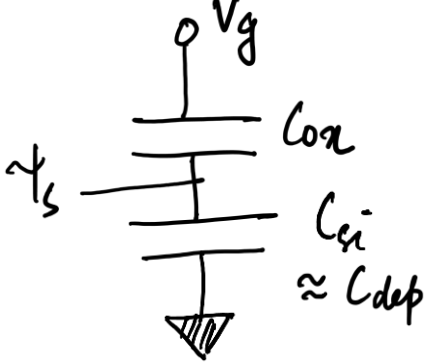


Including 2 terms Error < 0.1%!

$$I_{ds} = \frac{W}{L_m}\mu_{n,eff}C_{ox}\left[(V_{gs}-V_{tn})V_{ds} - \frac{1}{2}m V_{ds}^2\right] \quad (8)$$

Recall, $m \equiv$ BODY FACTOR

$$m = \frac{dV_{gs}}{d\psi_s} = \left(1+\frac{C_d}{C_{ox}}\right)$$



where $Q_{dep0} = \sqrt{2q\epsilon_{si}N_A(2\phi_{Bp})}$

& $C_{dep0} = \left.\frac{\partial Q_{dep0}}{\partial \psi_s}\right|_{\psi_s = 2\phi_{Bp}} = \sqrt{\frac{q\epsilon_{si}N_A}{4\phi_{Bp}}}$

Simpler Derivation of I_{ds} – Taylor Series Q_{inv}

$$Q_{inv}(y) = C_{ox}\left[V_{gs} - V_{fB} - (2\phi_{Bp} + V_y) - \frac{1}{C_{ox}}\sqrt{2q\epsilon_{si}N_A(2\phi_{Bp}+V_y)}\right] \Rightarrow (2\phi_{BP}+V_y)^{1/2} = \sqrt{2\phi_{BP}} + \frac{V_y}{2\sqrt{2\phi_{BP}}}$$ { Taylor series for $V_y \ll \phi_{BP}$ }

Substituting above :-

$$Q_{inv}(y) \approx C_{ox}\left[V_{gs} - V_{fB} - 2\phi_{BP} - \frac{\sqrt{2\epsilon_{si}N_A(2\phi_{BP})}}{C_{ox}} - \left(1 + \frac{1}{C_{ox}} \times \frac{1}{2}\sqrt{\frac{2q\epsilon_{si}N_A}{2\phi_{BP}}}\right)V_y\right]$$

$V_{fB} + 2\phi_{BP} + \frac{\sqrt{2\epsilon_{si}N_A(2\phi_{BP})}}{C_{ox}} = V_{th0}$

$$Q_{inv} = C_{ox}\left[V_{gs} - V_{th0} - m V_y\right]$$

C_{dep0}

$m \rightarrow$ body factor

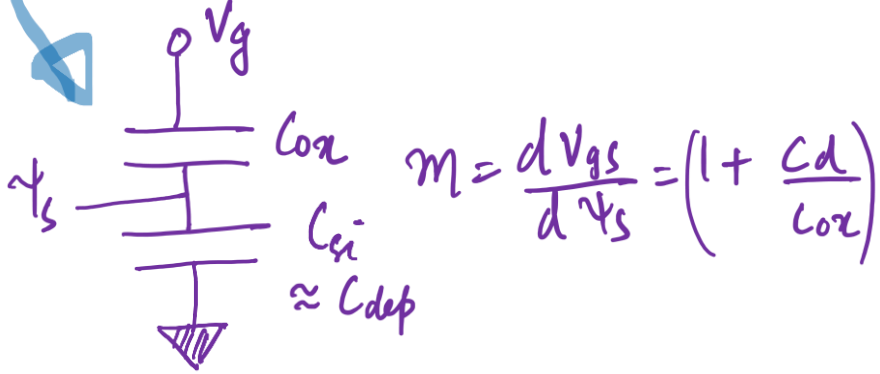
$$C_{dep_0} = \left.\frac{dQ_{si}}{d\psi_s}\right|_{\psi_s = 2\phi_B} = \frac{1}{2}\sqrt{\frac{2q\epsilon_{si}N_a}{2\phi_{BP}}}$$

$$I_{ds} = W \mu_{n,eff} Q_{inv}(y) \frac{dV_y}{dy}$$

$$I_{ds}\,dy = W \mu_{n,eff} Q_{inv}(y)\,dV_y$$

$$I_{ds}\int_0^{L_{ch}} dy = W \mu_{n,eff}\int_0^{V_{ds}} C_{ox}\left[V_{gs} - V_{th0} - m V_y\right]dV_y$$

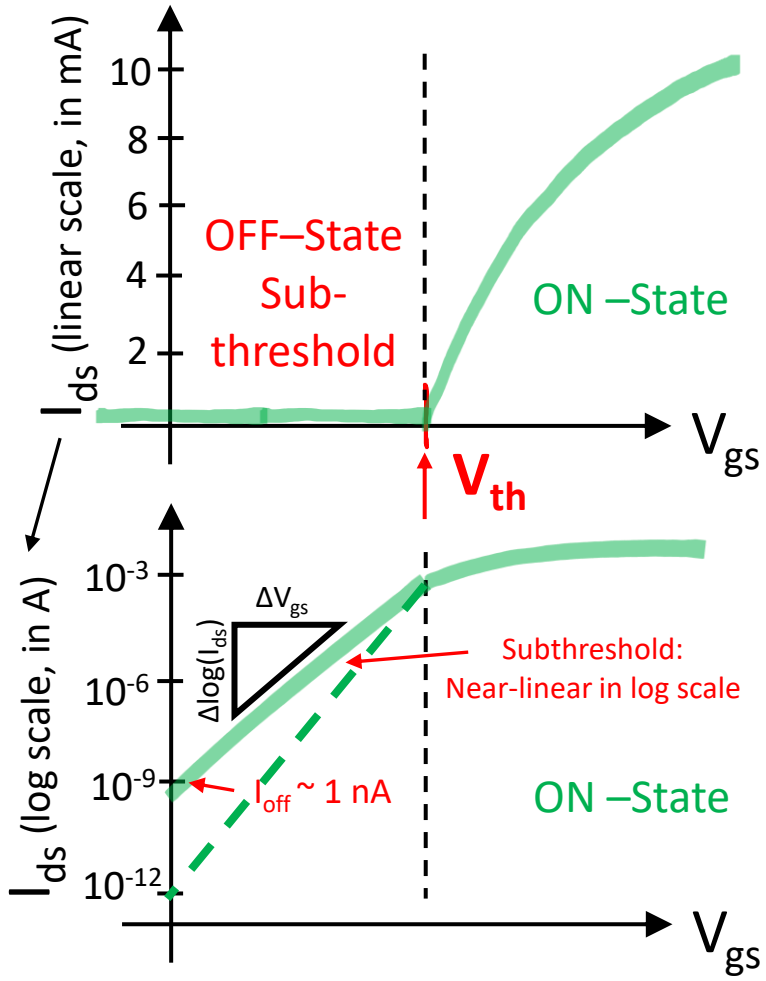
$$I_{ds} = \frac{W}{L_{ch}}\mu_{n,eff} C_{ox}\left\{(V_{gs} - V_{th0})V_{ds} - \frac{m V_{ds}^2}{2}\right\} \quad (8)$$



$$m = \frac{dV_{gs}}{d\psi_s} = \left(1 + \frac{C_d}{C_{ox}}\right)$$

Appendix -2 Complete SS derivation

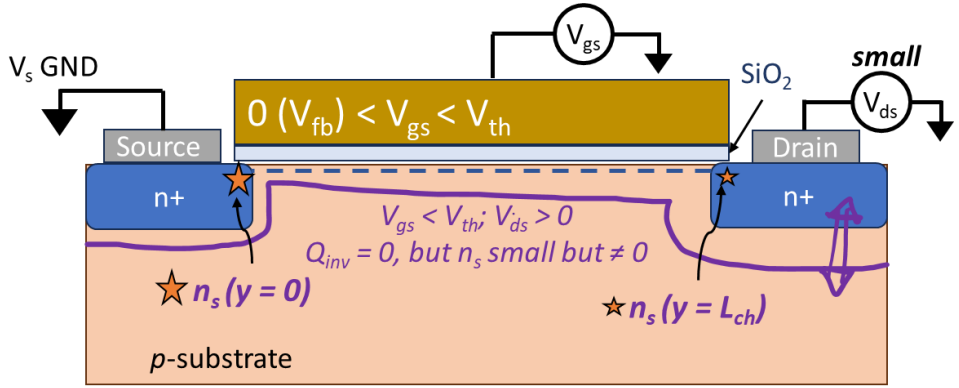
Till Now: I_{ds} ($V_{gs}>V_{th}$)
What happens I_{ds} ($V_{gs}< V_{th}$), $Q_{inv} = 0$

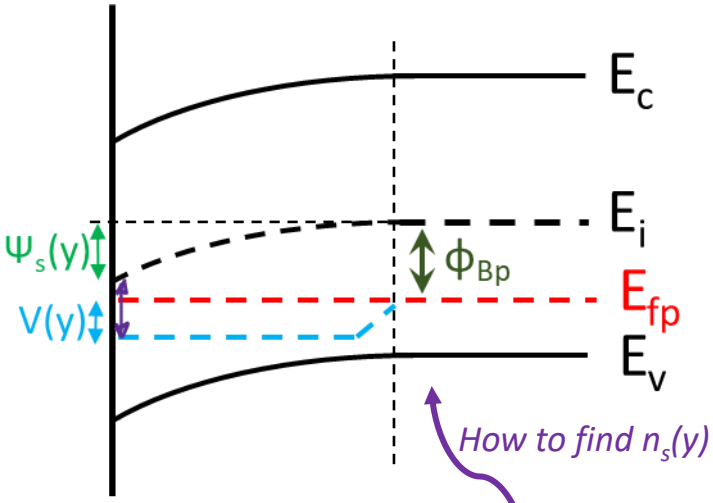


Subthreshold Swing, $SS = \frac{dV_{gs}}{d(log_{10} I_{ds})}$, in mV/dec

SS – Critical Parameter – Why? Steeper SS -> lower I_{off}
I_{off} - Leakage Power billions of MOSFETs in modern ICs

What is nature of SS conduction?





Since $n_s(y)$ very small $I_{n,drift} \approx 0$ Q: Why
But! Notice $\nabla n_s(y)$ due to V_{ds} (V_y): V_{rev}

I_{ds} ($V_{gs}< V_{th}$) due to Diffusion of n_s S->D! ⇒ $I_{ds}^{n,diff} = -qA D_n \frac{d n_s(y)}{dy}$

$$n_s(y) = n_i \exp\left[\frac{-\{E_i(x=0) - E_{fn}(x=0)\}}{V_T}\right] = n_i \exp\left[-\frac{\xi(\phi_{Bp} - \psi_s(y)) - V_y}{V_T}\right]$$

$$n_s(y) = n_{po} \exp\left[\frac{\psi_s(y) - V_y}{V_T}\right] = n_{po} \exp\left[\frac{\psi_s - V_y}{V_T}\right] \quad (9) \quad \{\psi_s(y) \approx \psi_s \text{ for } V_{gs} < V_{th}\}$$

Appendix -2 Complete SS derivation

Calculating I_{ds} with $n_s(y)$

$$I_{ds}^{n,diff} = -qAD_n \frac{dn_s(y)}{dy}$$

$$\int_0^{L_{ch}} I_{ds}\, dy = -qAD_n \int_{n_s(0)}^{n_s(L_{ch})} dn_s(y)$$

$$I_{ds} = \frac{qAD_n}{L_{ch}} [n_s(0) - n_s(L_{ch})]$$

$$n_s(0) = n_{po} \exp\left(\frac{\psi_s}{V_T}\right); \; n_s(L_{ch}) = n_{po} \exp\left(\frac{\psi_s - V_{ds}}{V_T}\right)$$

$$I_{ds} = \frac{qAD_n}{L_{ch}} n_{po} \exp\left(\frac{\psi_s}{V_T}\right)\left[1 - \exp\left(\frac{-V_{ds}}{V_T}\right)\right]$$

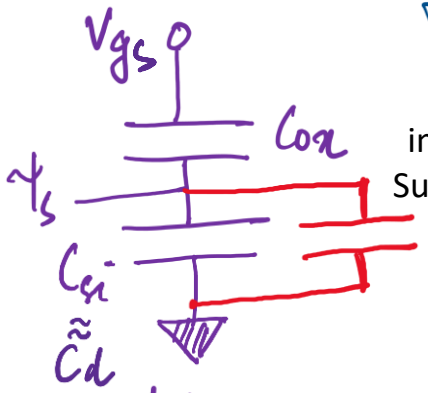
0 for $V_{ds} > 3V_T$

$$I_{ds}^{SS} \approx \frac{qAD_n}{L_{ch}} n_{po} \exp\left(\frac{\psi_s}{V_T}\right) \quad (10)$$

$$SS = \frac{\partial V_{gs}}{\partial \log(I_{ds})} = \frac{\partial V_{gs}}{\partial \psi_s} \frac{\partial \psi_s}{\partial \log(I_{ds})}$$

① ②

We are already know ① = m {Body factor}



Why C_{it} important in Subthreshold?

$$m = \frac{dV_{gs}}{d\psi_s} = \left(1 + \frac{C_d + C_{it}}{C_{ox}}\right)$$

To obtain ② use (10)

$$\log I_{ds} = \log[const] + \frac{1}{\ln(10)} \frac{\psi_s}{V_T}$$

$$\frac{\partial \psi_s}{\partial(\log I_{ds})} = \ln(10) V_T \approx 60\,mV \text{ @ } 300\,K$$

$2.3 \times 25.9 \approx 60$

$$SS = \frac{\partial V_{gs}}{\partial \log(I_{ds})} = \ln(10) V_T \left(1 + \frac{C_d + C_{it}}{C_{ox}}\right)$$

$SS \geq 60\,mV/dec$ } BOLTZMANN'S LIMIT of SS

I_{off} AND V_{dd} scaling $V_{dd} = V_{ss} + V_{ov} \sim 0.9$ V

Q: SS < 60 mV/dec? Hot Topic!

Sub-thermionic conduction

